

Phase 4–5 Research Direction Report

Prior-Art Validation • Novel Contribution • Patent/IP Position • Publication Strategy

RESEARCH-ONLY DECISION DOCUMENT

This report defines what the project is going to research, what will be published, what is only a patent/IP possibility, and what must be experimentally validated before any stronger claim is made.

Status: Phase 5 complete • Ready for Phase 6 engineering specification

1. Executive Decision

Phase 4 and Phase 5 have converted DAIL from a promising research idea into a defined research direction with a controlled novelty position, an experimental agenda, and a publication/IP strategy. The project is now ready to move into implementation and empirical validation, but DAIL is still a research hypothesis rather than a proven superior or patentable invention.

DECISION: Keep CloudShield AI as the broader product concept, keep DAIL as the research core, and proceed to Phase 6 only for engineering specification and implementation planning. Do not claim that DAIL is already proven novel or patentable.

2. What This Report Covers

This document intentionally covers only the research decisions produced in Phase 4 and Phase 5. Product/UI planning, general CloudShield features, and implementation details that do not affect the research claim are excluded.

- Exact research contribution after prior-art challenge.
- Academic, commercial, and patent overlap conclusions.
- Final novelty boundary and wording.
- Research questions and hypotheses that will be experimentally tested.
- Publication strategy and paper positioning.
- Patent/IP position and disclosure sequencing.
- Research reproducibility and evidence requirements.
- Exact statement of what we are doing next.

3. Phase 4 — What We Learned From Prior-Art Validation

Phase 4 was an adversarial validation phase. Instead of asking whether DAIL sounded interesting, we asked whether the exact mechanism was already present in academic work, commercial products, or patents.

3.1 Academic prior-art conclusion

The academic search showed that the individual building blocks of DAIL are already well represented. LLM-based Terraform remediation, layered verification, verifier feedback, formal contracts/invariants, dependency-aware or incremental verification, and security-regression analysis all have relevant prior art.

Area	Conclusion
LLM Terraform remediation	Established; not a novelty claim.
Layered verification/deceptive-fix detection	Established; TerraProbe provides strong direct evidence.
Verifier-feedback repair loops	Established; verifier-first Terraform work provides direct/adjacent evidence.
Regression in iterative LLM-IaC repair	Established problem; recent work directly measures it.
Invariant/contract refinement	Established in adjacent software-verification work such as ConVer/PolyVer.
Incremental/selective verification	Established in adjacent formal-methods literature.
AWS formal reasoning	Established by foundational systems such as Zelkova/Tiros.

The key research conclusion is therefore not that the regression problem itself is undiscovered. The research opportunity is the specific control mechanism we propose for preventing a newly generated Terraform state from silently discarding previously verified properties.

3.2 Commercial competitor conclusion

Commercial systems already cover much of the broad CloudShield product territory: cloud posture management, IaC scanning, risk context, dependency/impact analysis, blast-radius analysis, code-to-cloud mapping, and AI-assisted remediation. Therefore, generic “AI-powered CSPM” is definitively not the research contribution.

Commercial differentiation is not the research story. CloudShield may contain those capabilities as a product, but the paper must center on DAIL and its experimentally testable preservation/state-promotion mechanism.

3.3 Patent landscape conclusion

The patent landscape is crowded around AI/cloud remediation, IaC security, resource identification, dependency graphs, configuration correction, drift remediation, and LLM-assisted cloud remediation. The Phase 4 claim-level review did not identify a reviewed independent claim that clearly recites the complete DAIL sequence, but that is not a legal patentability conclusion.

- Individual DAIL components are not safe standalone patent-novelty claims.
- The strongest candidate IP focus is the interaction between protected verified-property state, infrastructure transition analysis, affected-property re-establishment, and trusted-state promotion.
- Novelty and inventive step/non-obviousness remain open and require professional claim analysis before filing.
- No patent filing decision is being made by this report.

4. Final Novelty Position

The final novelty position is intentionally narrow. DAIL is not being presented as a new primitive for AI remediation, dependency graphs, resource identity, invariant checking, or incremental verification.

CORE RESEARCH POSITION:

DAIL investigates a Terraform-specific state-promotion mechanism for iterative LLM remediation in which previously verified security and functional properties are carried forward as protection obligations, infrastructure changes determine which obligations may have been affected, and a candidate state is promoted to trusted status only after the affected obligations are re-established.

4.1 What is explicitly NOT claimed as novel

- AI-powered CSPM.
- LLM-based Terraform remediation.
- Terraform dependency analysis or blast-radius analysis.
- Terraform resource identity/correspondence.
- Invariants or contracts as a general concept.
- Incremental or selective verification as a general technique.

- Regression detection as a general concept.
- LLM-assisted formal verification as a general concept.
- Use of Z3/SMT itself.

4.2 What remains the candidate contribution

1. Previously verified infrastructure properties are treated as persistent protection obligations rather than passive historical records.
2. Terraform resource and dependency changes determine which protection obligations may no longer be safely trusted.
3. Affected obligations are re-established against the candidate infrastructure state.
4. The candidate state is not promoted to trusted status merely because the new security finding is fixed.
5. Promotion occurs only after the required preservation conditions are satisfied.

5. Final Research Model of DAIL

The core research object is the transition from a trusted infrastructure state to a new candidate state.

Trusted State → AI Candidate State → Change/Identity Analysis → Affected Protection Obligations → Re-verification → PROMOTE or REJECT

5.1 DAIL components

Component	Research role
Protected invariant state	Stores explicit previously verified security/functional properties as protection obligations.
Resource identity	Ensures obligations remain associated with the correct infrastructure objects and handles uncertainty conservatively.
Dependency analysis	Determines which protected properties may be affected by a change.
Invalidation / re-establishment	Removes unconditional trust from affected properties and requires them to be checked again.
Verification engine	Independently evaluates the new requirement and affected protection obligations.
Promotion gate	Decides whether the candidate state can become trusted.
Feedback loop	Returns evidence-based rejection information to the LLM for another bounded repair attempt.

6. Research Questions and Hypotheses

The research is now framed as a falsifiable empirical study rather than a design claim.

ID	Research question / hypothesis
RQ1	Can a stateful, dependency-aware state-promotion mechanism reduce unintended security and infrastructure regressions during iterative LLM-based Terraform remediation compared with stateless and full-reverification strategies?
RQ2	Does explicit protected-invariant state reduce regression rates compared with stateless verification?
RQ3	Does resource-identity-aware preservation reduce restructuring-related regressions?
RQ4	Can dependency-aware selective re-verification reduce verification work without causing an unacceptable increase in missed regressions?
RQ5	What safety, utility, latency, iteration, and implementation trade-offs does the preservation mechanism introduce?
H1	DAIL reduces unintended regression rate relative to stateless verification.
H2	DAIL reduces verification work relative to stateful full re-verification without an unacceptable increase in missed regressions.
H3	Resource-identity-aware preservation reduces restructuring-related regressions.
H4	DAIL prevents promotion of candidate states that violate modeled protected invariants.

7. What We Are Going to Do Experimentally

The primary experiment compares three conditions under the same Terraform scenarios, model configuration, and verification environment.

Condition	Protected state	Verification strategy	Purpose
A — Stateless	No	Current-state verification	Establish a realistic stateless baseline.
B — Stateful Full	Yes	Re-check all protected invariants	Isolate the value of stateful preservation without selective optimization.
C — DAIL	Yes	Dependency-aware re-verification of affected obligations	Test the complete proposed mechanism.

7.1 Primary research outcome

Unintended regression rate. A regression is defined as a previously protected invariant becoming false after an accepted remediation transition without an explicit intentional change to that invariant.

7.2 Secondary measurements

- Missed-regression rate.
- False-rejection rate.
- Repair success rate.
- Verification work / number of invariant checks.
- Protection coverage.
- Verification selectivity.
- Resource churn / restructuring.
- Iterations to convergence.
- Verification latency.
- LLM call/token usage where available.

7.3 Experimental scope

Dimension	Current scope
Cloud	AWS
IaC	Terraform
Core resources	VPC, EC2, Security Groups, RDS
Property types	Selected security and functional invariants
Task	Iterative LLM-based security remediation
Primary goal	Regression prevention and safe state promotion

8. What We Are Going to Publish

The intended publication is a research paper, not a general product brochure. The paper should be positioned as an empirical systems/software-engineering study with a cloud/IaC-security application.

PAPER IDENTITY:
Safe iterative LLM-based Terraform remediation with stateful invariant-aware state promotion.

8.1 Working contribution structure

- 6. A Terraform-specific state-promotion and invariant-preservation mechanism for iterative LLM remediation.
- 7. A resource/dependency-aware method for identifying protected properties that require re-establishment after a change.
- 8. A controlled evaluation comparing stateless, stateful-full, and DAIL strategies.
- 9. An empirical analysis of the safety–verification-efficiency trade-off.

8.2 Planned paper structure

- 10. Introduction and motivation.
- 11. Background: Terraform, LLM remediation, invariants and iterative verification.
- 12. Related work and prior-art boundary.
- 13. Research gap and formal problem definition.
- 14. DAIL design and state-promotion model.
- 15. Implementation for the selected Terraform/AWS subset.
- 16. Experimental methodology and baselines.
- 17. Results.
- 18. Discussion and interpretation.
- 19. Threats to validity and limitations.
- 20. Conclusion.

9. What About a Patent?

Patent protection remains an optional research/IP path, not an accomplished outcome. The current position is that the surrounding patent landscape is crowded, while the exact DAIL combination was not identified in the reviewed claim set.

Item	Current decision
Patentability	Not established.
Potential IP center	Trusted-state promotion controlled by re-establishment of affected previously verified infrastructure properties.
Immediate filing	No immediate filing decision.
Professional review	Required before making a filing or final legal claim.
Public disclosure	Potentially patent-sensitive details should remain controlled until the IP decision.
Indian CRI context	A later professional review should consider the current Indian CRI examination framework.

WIPO identifies novelty, inventive step/non-obviousness, industrial applicability and patent-eligible subject matter as key patentability considerations. WIPO also warns that public disclosure before filing can destroy novelty in many jurisdictions, while exceptions vary by jurisdiction. (WIPO, Patents FAQ; WIPO, How to Protect Inventions Through Patents).

For India, the Indian Patent Office currently lists the 2025 Guidelines for Examination of Computer Related Inventions (CRIs). The official guidance is relevant to any later patent-professional assessment of a computer-implemented DAIL mechanism. (Indian Patent Office, Guidelines for Examination of Computer Related Inventions (CRIs), 2025).

10. Publication and IP Timing — What We Will Do

The project should sequence technical development, IP assessment and public disclosure deliberately.

21. Keep the exact unpublished implementation under controlled/private development.
22. Build the prototype and determine the strongest technical mechanism actually produced.
23. Run the controlled experiments and identify the measured technical effect/trade-off.
24. Update the prior-art/claim comparison against the implemented mechanism.
25. Obtain professional IP guidance and decide whether any filing is justified.
26. Only then finalize public paper/code disclosure in line with the IP decision.

Important: this sequence is a research/IP management strategy, not legal advice. Public disclosure rules vary by jurisdiction and the final decision should be made with the institution/patent professional.

11. Research Evidence and Reproducibility Requirements

Every reported result should be traceable from the paper back to the exact experiment that produced it.

- Version the Terraform scenarios and invariant definitions.
- Record LLM model/version and generation configuration.
- Version prompts and DAIL code.
- Store trusted state, candidate patch, changed resources, affected invariants, verification evidence and final decision for every run.
- Keep raw results separate from processed analysis data.
- Generate final tables and figures from analysis scripts rather than manual transcription.
- Record Terraform/provider/tool versions and execution environment.
- Do not publish credentials, secrets or sensitive cloud configuration.
- Preserve an auditable experiment registry so every result has a run lineage.

12. What Is Still NOT Decided

- The exact final verifier stack (Z3/SMT is optional, not mandatory).
- The final invariant language/encoding.
- The exact stable-resource-identity algorithm for the selected Terraform subset.
- The final dependency/invalidation algorithm.
- The exact benchmark size and number of repeated LLM runs.
- The final publication venue.
- Whether an actual patent filing will be pursued.
- The ultimate empirical outcome of the DAIL hypotheses.

13. Research Claims We Must Avoid

- “DAIL guarantees cloud security.”
- “DAIL eliminates all Terraform regressions.”
- “DAIL is the first system to use dependency analysis/invariants/selective verification.”
- “DAIL is definitely patentable.”
- “DAIL is already proven better than full verification.”
- “Our verifier proves real-world AWS behavior beyond its explicitly modeled semantics.”

The research should instead use evidence-bounded language such as: “we propose,” “we investigate,” “within the prior art examined,” and, after experiments, “in the evaluated AWS/Terraform scope.”

14. Exact Research Direction Going Forward

WE ARE NOT BUILDING “ANOTHER AI CSPM.”

We are building and evaluating a narrowly scoped research prototype of DAIL inside CloudShield AI.

27. Define the exact DAIL software requirements from the locked research model.
28. Implement the protected invariant state and trusted/candidate state model.
29. Implement resource identity and dependency impact for the selected Terraform subset.
30. Implement invariant invalidation/re-establishment and the promotion gate.
31. Build the stateless and stateful-full baselines using the same LLM/verification environment.
32. Create controlled multi-step Terraform remediation scenarios.
33. Run repeated experiments and collect complete audit records.
34. Analyze regression, safety, verification-work and repair-utility outcomes.
35. Use the results to confirm, refine or reject the current DAIL research claim.
36. Only after the technical mechanism and evidence stabilize, finalize the publication/IP decision.

15. Final Research Decision

FINAL DECISION:

The research direction is DAIL. The paper will be about DAIL and its empirical evaluation. Patent protection is a separate, optional IP path that remains undecided until the implemented mechanism and results are reviewed professionally.

In one sentence:

“We are investigating whether preserving previously verified Terraform security and functional properties as stateful protection obligations can make iterative LLM-based remediation safer, while using resource/dependency impact to limit unnecessary re-verification.”

16. Evidence Basis and Key References

This report is based on the project’s Phase 4–5 research work plus the authoritative Phase 1/2 evidence artifacts that Phase 4–5 relied upon. The Phase 2 evidence explicitly states that its gap wording is an evidence-boundary statement, not a “first ever” claim.

- **Project evidence artifacts:**
CloudShield_AI_Phase_1_2_Faculty_Progress_and_Research_Direction_Report.docx;
CloudShield_AI_Phase_2_Consolidated_SLR_Report_v1.0.docx; Evidence boundary.md.
- **TerraRepair:** Mengistu et al., 2026, Tool-Grounded LLM Agent for Infrastructure-as-Code Repair.
- **TerraProbe:** Alsaid et al., 2026, Layered-Oracle Framework for Detecting Deceptive Fixes in LLM-Assisted Terraform Security Repair.

- **Verifier-First:** Jouini, 2026, Verifier-First Evaluation of Agentic LLMs for Infrastructure-as-Code Generation.
- **ConVer:** Using Contracts and Loop Invariant Synthesis for Scalable Formal Software Verification.
- **PolyVer:** A Compositional Approach for Polyglot System Modeling and Verification.
- **Does Fixing Break Security?:** Agyekum & Santos, 2026, Empirical Study of Security Degradation in Iterative LLM-Driven Infrastructure-as-Code Repair.
- **WIPO:** Patents FAQ and How to Protect Inventions Through Patents.
- **IP India:** 2025 Guidelines for Examination of Computer Related Inventions (CRIs).